Texas A&M University
Department of Electrical and Computer Engineering

ECEN 474/704 – (Analog) VLSI Circuit Design

Spring 2016

Exam #1

Instructor: Sam Palermo

- Please write your name in the space provided below
- Please verify that there are **5 pages** in your exam
- You may use one double-sided page of notes and equations for the exam
- Good Luck!

Problem	Score	Max Score
1		40
2		40
3		20
Total		**100**

Name: _____

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Problem 1 (40 points)

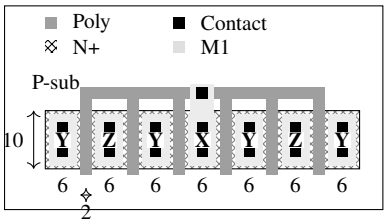
For the layout below, assume that all the commonly labeled diffusion areas are connected with the appropriate metal layers. Assume that $V_X = 1.2\,\text{V}$, $V_Y = 0\,\text{V}$, $V_Z = 0.4\,\text{V}$, $V_{T0} = 0.7\,\text{V}$, $\gamma = 0$, and that all Spice parameters are given (i.e. C_j, C_{jsw}, C_{jc}, C_{ox}, C_{ov}). The dimensions are given in μm, with all the poly gates having a length of $2\,\mu$m and $L_D = 0.1\,\mu$m.

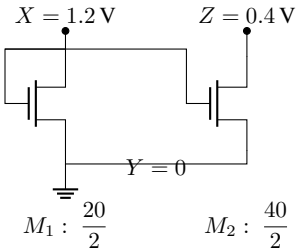
a) Draw the equivalent circuit. Combine all parallel transistors and give the total width and length of the equivalent transistors. (15 points)

b) What region(s) are the transistors operating in? (5 points)

c) For node **X only**, give an expression and calculate the total gate cap. (10 points)

d) For node **Z only**, give an expression and calculate the total junction cap. Note for the perimeter terms, include the sides underneath the gate. (10 points)





a. The six unit transistors each have $W/L = 10/2$. Two are connected from X to Y and four from Z to Y; all gates connect to X. Combining them gives the two-transistor circuit above.

b. $M_1:\ V_{GS} = V_{DS} = 1.2\,\text{V} > V_T$, and $V_{DS} > V_{GS} - V_T$: saturation.

$M_2:\ V_{GS} = 1.2\,\text{V}$, $V_{DS} = 0.4\,\text{V} < 1.2 - 0.7 = 0.5\,\text{V}$: triode.

c. With $L_{\text{eff}} = 2 - 2(0.1) = 1.8\,\mu$m, M_1's gate–drain capacitance is shorted out because gate and drain both connect to X. Thus

$$C_{gX,1} = C_{gs,1} = \frac{2}{3}(20\,\mu\text{m})(1.8\,\mu\text{m})C_{ox} + (20\,\mu\text{m})C_{ov},$$
$$C_{gX,2} = C_{gs,2} + C_{gd,2} = 2\left[\frac{1}{2}(40\,\mu\text{m})(1.8\,\mu\text{m})C_{ox} + (40\,\mu\text{m})C_{ov}\right].$$

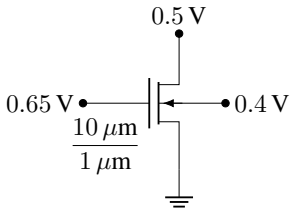
$$\boxed{C_{gX} = 96\,\mu\text{m}^2 C_{ox} + 100\,\mu\text{m}C_{ov}.}$$

d. The two Z diffusion regions have total area $A_Z = 2(6)(10) = 120\,\mu\text{m}^2$ and perimeter $P_Z = 2 \cdot 2(6+10) = 64\,\mu$m, counting the gate-adjacent sides. Following the supplied solution's capacitance notation,

$$C_{jZ} = A_Z C'_j + P_Z C'_{jsw} + C_{jc}\frac{W_2 L_{\text{eff}}}{2}$$
$$= 2(6)(10)\,\mu\text{m}^2 C'_j + 2 \cdot 2(6+10)\,\mu\text{m}C'_{jsw} + \frac{(40)(1.8)}{2}\,\mu\text{m}^2 C_{jc},$$
$$\boxed{C_{jZ} = 120\,\mu\text{m}^2 C'_j + 64\,\mu\text{m}C'_{jsw} + 36\,\mu\text{m}^2 C_{jc}.}$$

Problem 2 (40 points)

For the transistor below, assume $V_{T0} = 0.7\,\text{V}$, $\gamma = 0.45\,\text{V}^{1/2}$, and $2\Phi_F = 0.9\,\text{V}$.



$$I_{DS} = I_0 \exp\left(\frac{(V_{GS}-V_{Tn})q}{\zeta kT}\right)\left(1-\exp\left(-\frac{V_{DS}q}{kT}\right)\right) \quad \text{(Subthreshold)}$$

$$I_{DS} = B\frac{W}{L-2L_D}(V_{GS}-V_{Tn})^{\alpha}(1+\lambda V_{DS})\left(2-\frac{V_{DS}}{V_{DSAT}}\right)\frac{V_{DS}}{V_{DSAT}} \quad \text{(Triode)}$$

$$I_{DS} = B\frac{W}{L-2L_D}(V_{GS}-V_{Tn})^{\alpha}(1+\lambda V_{DS}) \quad \text{(Saturation)}$$

$$V_{Tn} = V_{T0} + \gamma\left(\sqrt{|2\Phi_F| + V_{SB}} - \sqrt{|2\Phi_F|}\right), \qquad V_{DSAT} = V_{GS} - V_{Tn}.$$

a. Calculate V_T and state the transistor's operation region. (10 points)

$$V_T = 0.7 + 0.45\left(\sqrt{0.9-0.4} - \sqrt{0.9}\right) = \boxed{0.591\,\text{V}},$$

$$V_{GS} - V_T = 0.65 - 0.5913 = 58.7\,\text{mV}, \qquad V_{DS} = 0.5\,\text{V} > V_{GS} - V_T.$$

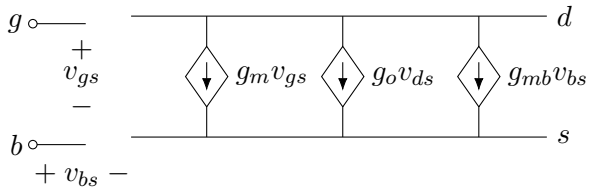
Operation region = $\boxed{\text{Saturation}}$.

b. Using the appropriate I_{DS} equation above, derive and sketch the small-signal model of the transistor. Include expressions for g_m, g_o, and g_{mb}. (20 points)

$$g_m = \left.\frac{\partial I_{DS}}{\partial V_{GS}}\right|_Q = B\frac{W}{L_{\text{eff}}}\alpha(V_{GS}-V_T)^{\alpha-1}(1+\lambda V_{DS}),$$

$$g_o = \left.\frac{\partial I_{DS}}{\partial V_{DS}}\right|_Q = B\frac{W}{L_{\text{eff}}}(V_{GS}-V_T)^{\alpha}\lambda,$$

$$g_{mb} = \left.\frac{\partial I_{DS}}{\partial V_{BS}}\right|_Q = \boxed{\frac{\gamma g_m}{2\sqrt{|2\Phi_F| + V_{SB}}}}.$$



c. Assume that $B = 65\,\mu\text{A/V}^{\alpha}$, $\lambda = 0.1\,\text{V}^{-1}$, and $L_D = 0.1\,\mu\text{m}$. Calculate the transistor's intrinsic gain (g_m/g_o) for both $\alpha = 2$ and $\alpha = 1$. (10 points)

$$\frac{g_m}{g_o} = \frac{\alpha}{V_{GS}-V_T}\frac{1+\lambda V_{DS}}{\lambda}.$$

$$\alpha = 2: \quad \frac{2}{58.7\,\text{mV}}\frac{1+(0.1)(0.5)}{0.1\,\text{V}^{-1}} \simeq \boxed{358\,\text{V/V}}, \qquad \alpha = 1: \quad \boxed{179\,\text{V/V}}.$$

Problem 3 (20 points)

Sketch a layout that matches three capacitors of unit size 2.2, 4.5, and 6.8. Assume that the unit capacitors are sized $5\,\mu\text{m} \times 5\,\mu\text{m}$. Make sure to give the non-unit capacitor(s) dimensions. In the sketch clearly label the critical dimensions and use at least 2 layout matching techniques. Also, write specifically the 2 layout matching techniques that you are using. Note, if you use a common-centroid layout technique, it doesn't have to have perfect center-of-mass matching, but it should be close.

Need to match perimeter/area for all capacitors. A unit capacitor has $A_u = 25\,\mu\text{m}^2$ and $P_u/A_u = 4/5\,\mu\text{m}^{-1}$.

$$\begin{aligned} C_1 = 2.2C_u: &\quad 1 \text{ unit cap} + 1 \text{ non-unit cap}, &\quad A_{nu,1} = 1.2A_u, \\ C_2 = 4.5C_u: &\quad 3 \text{ unit caps} + 1 \text{ non-unit cap}, &\quad A_{nu,2} = 1.5A_u, \\ C_3 = 6.8C_u: &\quad 5 \text{ unit caps} + 1 \text{ non-unit cap}, &\quad A_{nu,3} = 1.8A_u. \end{aligned}$$

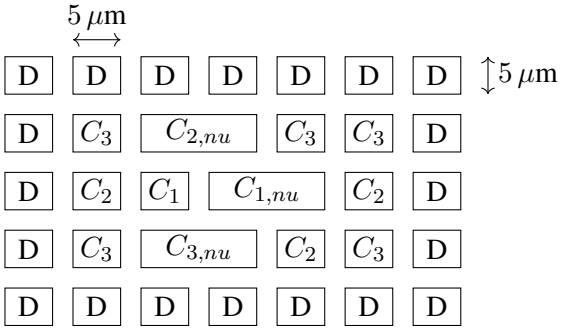
For a non-unit capacitor of area kA_u, equal perimeter-to-area ratio gives

$$x_{nu}y_{nu} = kx_u^2, \qquad \frac{2(x_{nu}+y_{nu})}{x_{nu}y_{nu}} = \frac{4}{x_u}, \qquad x_u = 5\,\mu\text{m}.$$

Thus, taking y_{nu} to be the shorter side,

$$y_{nu} = x_u\left(k - \sqrt{k^2-k}\right), \qquad x_{nu} = \frac{kx_u^2}{y_{nu}}.$$

Capacitor	k	y_{nu}	x_{nu}
C_1	1.2	$3.55\,\mu$m	$8.45\,\mu$m
C_2	1.5	$3.17\,\mu$m	$11.83\,\mu$m
C_3	1.8	$3.00\,\mu$m	$15.00\,\mu$m



The non-unit rectangles have the dimensions tabulated above; the placement sketch is not to scale. Capacitors with the same C_i label are connected in parallel. D denotes a dummy capacitor.

Layout Technique #1 = Dummies around the entire perimeter.

Layout Technique #2 = Approximately common-centroid placement.

Scratch Paper